

INTRODUCTION

This report outlines the research and initial work performed on a sensing and control system for a robot. Research into existing technologies is outlined, with appropriate references. The testing performed to validate the technologies used is described. The overall design, at the time of writing, is described, along with the functional description of each module. The robot is being

developed by a team of three bachelor of computer application Students, of which the author is one. Each member of the team has been assigned a specific role within the project. The team members are responsible for individually documenting their progress. Herbie (humanoid experimental robot bio-type integrated with electronic)

Our team have created a humanoid robot which is used as a healthcare companion. The robot is used to sense the user's body temperature, pressure and heartbeat the robot will act as a personal assistant. Our system provides security to access it. The robot can move using its rotating Wheels. It can be controlled using Bluetooth application, voice command , and IR remote mode for communication The robot will access the temperature and blood pressure and blood oxygen level through max300 Sensor and dht11 temperature sensor.

PURPOSE OF THE PROJECT

The basic concept behind this project is to introduce a humanoid healthcare companion robot which can also act as a personal assistant. Robotics have increased productivity and resource efficiency in the industrial and retail sectors, and now there is an emerging interest in realizing a comparable transformation in healthcare. Robotics and artificial intelligence (AI) are some of the latest promising technologies expected to increase the quality and safety of care while simultaneously restraining expenditure and, recently, reducing human contact too. Healthcare robots are likely to be deployed at an unprecedented rate due to their reduced cost and increasing capabilities such as carrying out medical interventions, supporting biomedical research and clinical

SCOPE OF THE PROJECT

Since the advent of the COVID-19 pandemic, the healthcare industry has been flooded with novel technologies to assist the delivery of care in unprecedented circumstances. Staff vacancy levels increased, social restrictions curtailed many traditional means of care delivery, and stringent infection control measures brought new challenges to human-delivered care. Although many of the challenges that the pandemic brought onto healthcare have subsided, staff burnout, an increasingly elderly population, and backlog strains caused by the pandemic have meant that staff shortages persist across healthcare systems across the world.

Robotic systems have long been cited to be able to alleviate workforce pressures, not least in healthcare. [11] Such systems can include remote presence robots for virtual consultations or transportation robots for automated delivery of equipment within hospitals.

SYSTEM ANALYSIS

System Analysis is concerned with analyzing, designing, implementing and evaluating information system in our organization. It is carried out to make the system more effective either by modification or by substantial redesign. In system analysis we identify the problem, study the alternative solution and select the most suitable solution, which meet the technical economic and social demands for analysis, various tools such as dataflow diagram, interviews on site observation, questionnaires etc., are used. System analysis process is also called a life cycle methodology since it relates to four significant phases in life cycle of all information system.

They are

1. System Analysis / Study Phase
2. System Design / Design phase.
3. System Development / Development Phase.
4. Testing and implementation / Operation Phase.

All activities associated with each life cycle phase must be performed managed and documented. So, system analysis is the performance, management and documentation of the activities related to the four life cycle phases of a computer-based system.

The basic concept behind the project “FITME” presents a novel method of collaborating ease in online shopping as well as for customer satisfaction while doing shopping online. And this model estimates human body measurements from human real-time pictures. as well as for customer satisfaction while doing shopping online.

STUDY PHASE

EXISISTING SYSTEM

In the recent past, health chatbots have been on the rise. As we see more bots emerging with incredible speed, we decided to take a closer look. We’ve gathered some of the industry leaders to get a general idea of where the health chatbot industry is heading.

One Remission

OneRemission is an oncology application augmented with a chatbot for those battling the disease and their loved ones. Cancer patients and survivors can access information on diets, lifestyle recommendations, stress management practices, etc. Family and caregivers can learn how they should behave and process their feelings.

The app is put together by licensed professionals and acts as an occasional substitute for a doctor’s visit. There is also a feature that connects you with an online oncologist, who is available at any time.

Youper

This application is aimed at monitoring and improving users’ emotional health. Youper’s AI combines mental health research and user data to match the right psychological techniques with users. To get a sense of what a user is going through, the app asks them how they are feeling from a menu of descriptive words. They can rate that emotion from “slightly” to “extremely.”

It also offers personalized meditations, mindfulness exercises, and journaling prompts. Overall, the app developers wanted to promote mental health awareness and encourage more people to seek professional help.

PROPOSED SYSTEM

In the existing system we can see that the above health care applications are focused only on software .But we focus not only on software but also hardware,for example we focus on hardware parts like temperature sensors,manual controled movement using mobile application which is developed by using MIT app inventor which can be very usefull for the user to handle the robotic hardware.another advantage of our project is that the user will get a real humanoid feel which is much better interacted with.another advantage of our robot is that we can predefine the route or pathway of our robot and can act as a line following robot using the app.

PROJECT SHEDULING

S NO	PHASES	START DATE	END DATE	NO.OF DAYS
1	Study phase	5-11-22	13-11-22	5
2	Designing phase	19-11-22	4-12-22	6
3	Development phase	10-12-22	25-12-22	6
4	Testing	28-1-23	29-1-23	2
5	Implementation	30-1-23	1-2-23	3
6	Chengannur IHRD tharang fest presentation	2-2-23	6-2-23	5
7	Baithul Izza arts and science college presentation	14-2-23	14-2-23	1
8	SNES college chathamangalam presentation	26-2-23	27-2-23	2

SYSTEM REQUIREMENT SPECIFICATION

Software specification

A software requirements specification (SRS) is a document that describes what the software will do and how it will be expected to perform. It also describes the functionality the product needs to fulfill all stakeholders.

• Operating system:

An operating system (OS) is a software system that manages computer hardware and software resources and provides common services for computer programs. It acts as a communication bridge between the computer hardware and software, and allows users to interact with the computer and run various applications.

An operating system performs a variety of functions, including managing computer memory, controlling input and output devices, managing file systems, providing security and protection, and scheduling tasks.

• Technology Used: python

Python is a high-level, interpreted programming language that was first released in 1991. It is designed to be easy to read and write, with a syntax that emphasizes code readability, and is widely used in a variety of applications, including web development, data analysis, machine learning, and scientific computing.

Python is known for its simplicity, readability, and ease of use. It has a large and active user community, which has contributed to a wide range of libraries and frameworks, making it a popular choice for both beginners and experienced developers. Some of the popular libraries and frameworks for Python include NumPy, Pandas, TensorFlow, Django, and Flask.

One of the key features of Python is its dynamic nature, which allows for easy prototyping and rapid development. It also has a strong focus on code readability, using indentation and whitespace to define code blocks, rather than curly braces and semicolons.

Python is an open-source language, which means that the source code is freely available for anyone to use, modify, and distribute. It is available on multiple platforms, including Windows, macOS, and Linux, and can be downloaded from the official Python website.

Python 3.7 is a specific version of the Python programming language, released in 2018. It is the latest version in the 3.x series and includes a number of new features and improvements over the previous versions.

Some of the notable features of Python 3.7 include:

Data classes: This is a new way to define classes that are primarily used to store data. It provides a concise syntax for defining classes that have a fixed set of attributes.

Improved type hints: Type hints are annotations that can be added to Python code to indicate the expected types of variables and function parameters. Python 3.7 includes improvements to the type hint syntax, making it easier to use and more expressive.

Performance improvements: Python 3.7 includes a number of performance improvements, including faster startup times and better memory management.

New built-in modules: Python 3.7 includes several new built-in modules, such as `contextvars`, which provides a way to share state between coroutines, and `dataclasses`, which provides a way to define classes for storing data.

Overall, Python 3.7 is a significant update to the language, with many new features and improvements that make it an attractive option for developers. It is widely used in a variety of applications, including web development, data science, and machine learning.

• **IDE: arduino ide**

The Arduino Integrated Development Environment (IDE) is a software application used for programming Arduino microcontroller boards. It provides a user-friendly interface for writing and uploading code to the Arduino board.

The Arduino IDE includes a code editor with syntax highlighting, auto-completion, and error checking features. It also includes a serial monitor for debugging and testing the communication between the computer and the Arduino board.

The IDE supports various programming languages, including C and C++. It provides a set of built-in libraries that make it easy to work with the hardware components of the Arduino board, such as digital and analog pins, sensors, and actuators.

The Arduino IDE is free and open-source software, which means that the source code is available for anyone to view, modify, and distribute. It is available for multiple platforms,

including Windows, macOS, and Linux, and can be downloaded from the official Arduino website.

Overall, the Arduino IDE is a powerful tool for programming Arduino boards, even for those with little or no programming experience. Its user-friendly interface and comprehensive set of libraries make it easy to get started with electronics projects.

MIT app inventor

MIT App Inventor is a web-based development platform for creating mobile applications for Android devices. It was developed by the Massachusetts Institute of Technology (MIT) and launched in 2010 as a free and open-source software.

MIT App Inventor allows users to create mobile apps without the need for extensive programming knowledge. It uses a drag-and-drop interface that allows users to add components and features to their apps using pre-built blocks of code. The platform also provides a visual interface for designing the user interface of the app.

Some of the features of MIT App Inventor include:

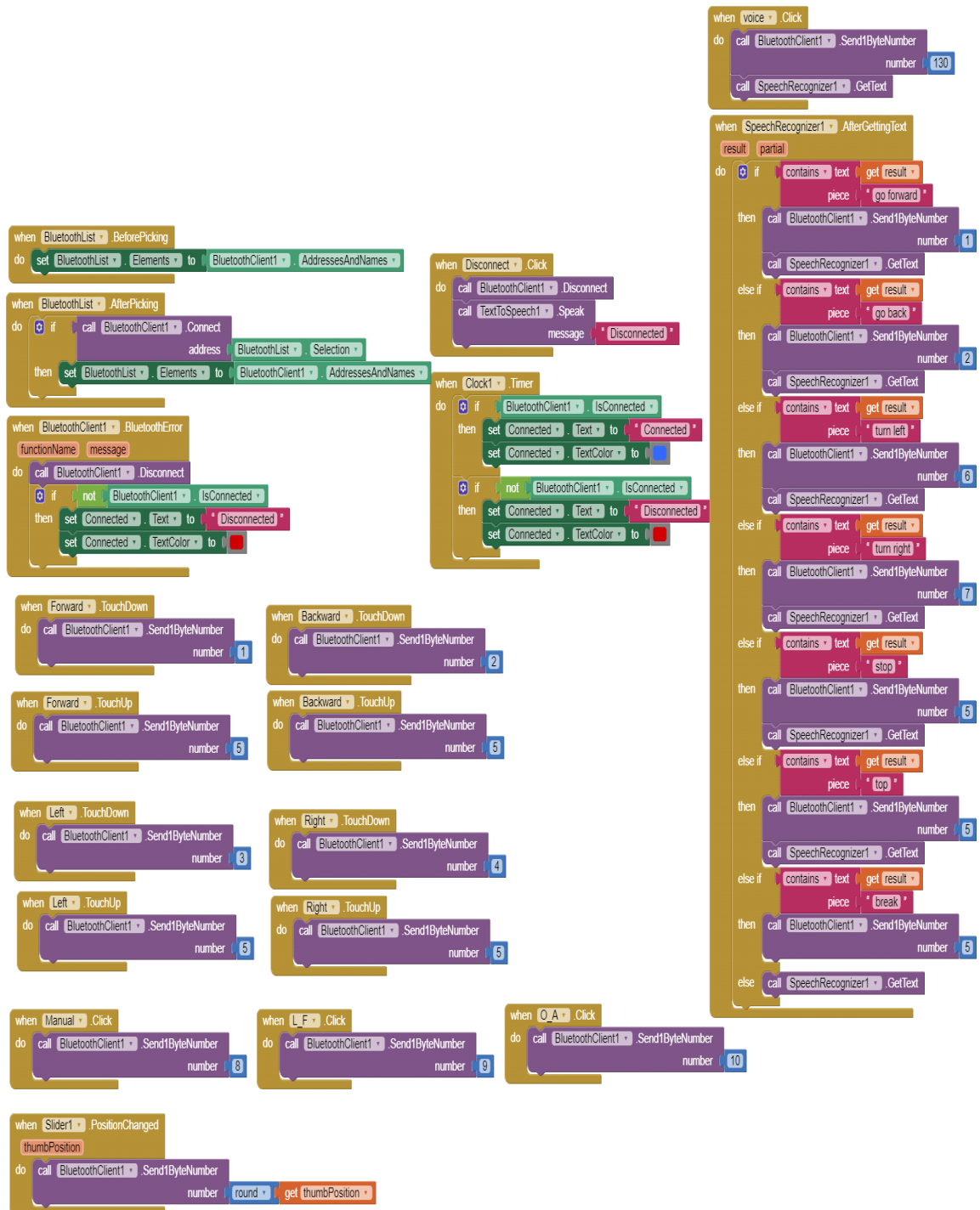
Easy to use: MIT App Inventor is designed to be easy to use, even for beginners with little or no programming experience.

Open-source: The platform is free and open-source, which means that the source code is available for anyone to view, modify, and distribute.

Wide range of components: The platform provides a wide range of components and features that can be added to an app, including sensors, multimedia, and social media integration.

Real-time testing: The platform allows users to test their app in real-time on their Android device, making it easy to see how the app will look and function.

Overall, MIT App Inventor is a powerful tool for creating mobile apps without the need for extensive programming knowledge. Its user-friendly interface and comprehensive set of features make it a popular choice for educators, students, and hobbyists.



HARDWARE SPECIFICATION

Arduino

Arduino (/ɑ:r'dwi:nou/) is an open-source hardware and software company project, and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices. Its hardware products are licensed under a CC BY-SA license, while the software is licensed GNU General Public License (GPL),[1] permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially from the official website or through authorized distributors.[2] Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards ('shields') or breadboards (for prototyping) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs. The microcontrollers can be programmed using the C and C++ programming languages, using a standard API which is also known as the Arduino Programming Language, inspired by the Processing language and used with a modified version of the Processing IDE. In addition to using traditional compiler toolchains, the Arduino project provides an integrated development environment (IDE) and a command line tool developed in Go.

Dht11 sensor

The digital temperature and humidity sensor DHT11 is a composite sensor that contains a calibrated digital signal output of temperature and humidity. The technology of a dedicated digital modules collection and the temperature and humidity sensing technology are applied to ensure that the product has high reliability and excellent long-term stability. The sensor includes a resistive sense of wet component and an NTC temperature measurement device, and is connected with a high-performance 8-bit microcontroller. The schematic diagram of the Humiture Sensor Module is as shown following: Only three pins are available for use: VCC, GND, and DATA. The communication process begins with the DATA line sending start signals to DHT11, and DHT11 receives the signals and returns an answer signal. Then the host receives the answer signal and begins to receive 40-bit humiture data (8-bit humidity integer + 8-bit humidity decimal + 8-bit temperature integer + 8-bit temperature decimal + 8-

bit checksum voltage regulator is a system designed to automatically maintain a constant voltage. A voltage regulator may use a simple feed-forward design or may include negative feedback. It may use an electromechanical mechanism, or electronic components. Depending on the design, it may be used to regulate one or more AC or DC voltages. 7805, 7806, 7809, 7812 voltage regulator Electronic voltage regulators are found in devices such as computer power supplies where they stabilize the DC voltages used by the processor and other elements. In automobile alternators and central power station generator plants, voltage regulators control the output of the plant. In an electric power distribution system, voltage regulators may be installed at a substation or along distribution lines so that all customers receive steady voltage independent of how much power is drawn from the line.

Minicomputer

The min computer is used for giving visual data in our 7 inch display module and for Herbie's voice assistant. We have used Microsoft windows 11 as operating system 12v12ah battery. 12-volt batteries usually consist of three cells connected in parallel. Each cell can produce around 4 volts of power, making up a total of 12 volts of electrical power – which is enough to power vehicles, RVs, home energy storage systems, and most household appliances.

IR sensor

A passive infrared sensor (PIR sensor) is an electronic sensor that measures infrared (IR) light radiating from objects in its field of view. They are most often used in PIR-based motion detectors. PIR sensors are commonly used in security alarms and automatic lighting applications Hc05 Bluetooth module HC-05 module is an easy to use Bluetooth SPP (Serial Port Protocol) module, designed for transparent wireless serial connection setup. Serial port Bluetooth module is fully qualified Bluetooth V2.0+EDR (Enhanced Data Rate) 3Mbps Modulation with complete 2.4GHz radio transceiver and baseband. Stepdown buck convertor A buck converter, also known as a step-down converter, is a DC/DC power converter that provides voltage step down and current step up. The converter reduces the voltage when the power source has a higher voltage than V_{in} .

L293D motor driver

The L293D is designed to provide bidirectional drive currents of up to 600-mA at voltages from 4.5 V to 36 V. Both devices are designed to drive inductive loads such as relays, solenoids, DC and bipolar stepping motors, as well as other high-current/high-voltage loads in positive-supply applications. 12v30rpm motor 30RPM 12V DC geared motors for robotics

applications. It gives a massive torque of 32Kgcm. The motor comes with metal gearbox and off-centered shaft. Features: 30RPM 12V DC motors with Metal gearbox and metal gears.

10a12v dc smart charger

battery charger, recharger, or simply charger,[1][2] is a device that stores energy in a battery by running an electric current through it. The charging protocol (how much voltage or current for how long, and what to do when charging is complete) depends on the size and type of the battery being charged

10*4 dimension tire

Often referred to as a 10 inch tire, this tire is one of the most common found on hand trucks, pressure washers, leaf blowers, air compressors, generators, garden carts and nursery carts. Also See: Understanding Tire Sizes.

360 freewheels

ID360 is a ratchet freehub system designed and manufactured by Mavic. Brought to road and gravel in 2016, this system was adopted for the first time on MTB wheels in 2018. ID360 is stiff and durable while benefiting from faster engagement.

Mg995 servo motor

MG995 servo is a simple, commonly used standard servo for your mechanical needs such as robotic head, robotic arm. It comes with a standard 3-pin power and control cable for easy using and metal gears for high torque. A Me RJ25 Adapter also help you to connect the servo with Me Baseboard or Makeblock Orion easily.

Microphone

It is used to convert sound waves into electric waves or input the audio into computers. It captures audio by converting sound waves into an electrical signal, which may be a digital or analog signal. This process can be implemented by a computer or other digital audio devices.

4 ohm 3 v speaker

The speaker has a resistance of 4 ohms and a power rating equal to 3W. Its small size allows it to be used in various applications where space is at a premium. This is used in general

warning alarms for some projects, Simple amplifier projects, For TV & Music systems, and will also fit in robotics projects

7 inch display

This 7 Inch Monitor Display Screen is the smallest possible HD display with high-resolution and high-contrast visibility. The 7 Inch Monitor Display Screen module designed especially for FPV and outdoor operation purposes.

IR receiver

An infrared receiver, or IR receiver, is hardware that sends information from an infrared remote control to another device by receiving and decoding signals. In general, the receiver outputs a code to uniquely identify the infrared signal that it receives.

Tecnology specification

python:Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together. Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages, which encourages program modularity and code reuse. The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributed.

android:Android is a mobile operating system based on a modified version of the Linux kernel and other open-source software, designed primarily for touchscreen mobile devices such as smartphones and tablets. Android is developed by a consortium of developers known as the Open Handset Alliance, though its most widely used version is primarily developed by Google

windows 10: it is a major release of Microsoft's Windows NT operating system. It is the direct successor to Windows 8.1, which was released nearly two years earlier. It was released to manufacturing on July 15, 2015, and later to retail on July 29, 2015.[18] Windows 10 was made available for download via MSDN and TechNet, as a free upgrade for retail copies of Windows 8 and Windows 8.1 users via the Windows Store, and to Windows 7 users via

Windows Update. Windows 10 receives new builds on an ongoing basis, which are available at no additional cost to users, in addition to additional test builds of Windows 10, which are available to Windows Insiders. Devices in enterprise environments can receive these updates at a slower pace, or use long-term support milestones that only receive critical updates, such as security patches, over their ten-year lifespan of extended support.[19][20] In June 2021, Microsoft announced that support for Windows 10 editions which are not in the Long-Term Servicing Channel (LTSC) will end on October 14, 2025.[16]

DESIGN PHASE

Introduction

This section examines the overall structure of the Sensors and Control System. The Hardware and Software aspects of the system are discussed in turn, with brief justifications for different aspects of the design.

Hardware design hierarchy

The laptop computer is the control center for the robot, and is the platform for the software needed to control the robot. The laptop has Wireless LAN capability, allowing the robot to be controlled from another computer on the same network. This prevents the decision system being constrained by the specifications of the laptop computer. The robot sensors and motors are interfaced using a USB/I2C conversion device. This allows commands to be sent to the sensor controller/motor controller through a standard serial port software interface. The motor controller and sensor controller reside on the I2C bus, as mentioned above. The MD22 is a self contained prefabricated unit, requiring only control and power inputs, whereas the sensor controller is a bespoke.

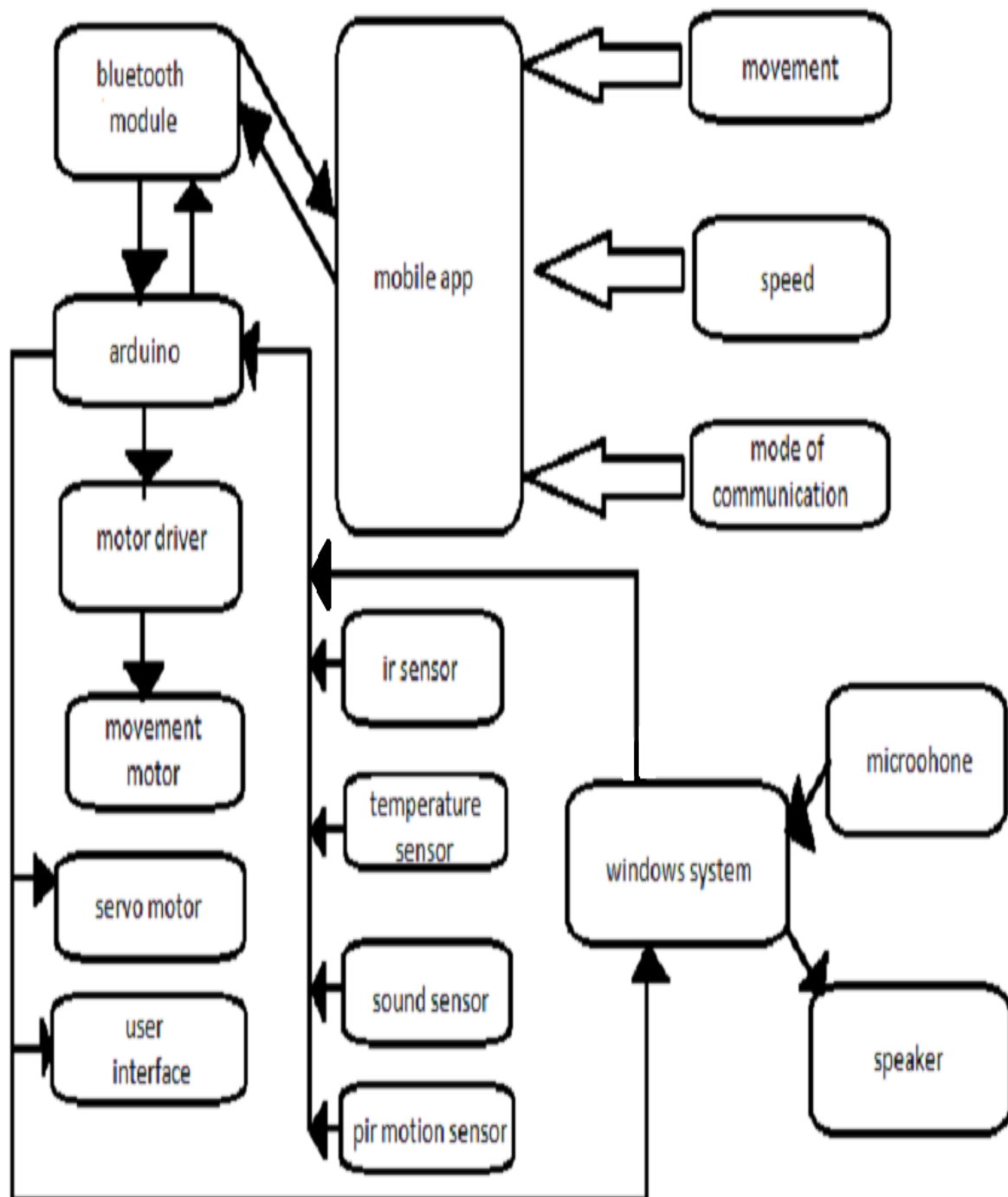
circuit consisting of:

- . Arduino uno
- . Mini computer
- . L293D motor driver
- . Tires (10*4 daimention)
- . 360 freewheels
- . 12v30rpm motor

- . Arduino mega
- . 7805 voltage regulator
- . 7806 voltage regulator
- . 7809 voltage regulator
- . 7812 voltage regulator
- . 12v12ah battery
- . Hc05 Bluetooth module
- . Stepdown buck convertor
- . 10a12v dc charger
- . Dht11 sensor
- . Mg995 servo motor
- . Microphone
- . 4ohm 3v speaker
- . Body filler
- . 7 inch display
- . IR receiver
- . USB connector
- . Toggle switch
- . 2 way switch
- . 1 way switch

model desing

CICUTE DESING
DATA FLOW DIAGRAM



CODING

CODING GUIDELINES

Writing an efficient software code requires a thorough knowledge of programming. This knowledge can be implemented by following a coding style which comprises several guidelines that help in writing the software code efficiently and with minimum errors. These guidelines, known as coding guidelines, are used to implement individual programming language constructs, comments, formatting, and so on. These guidelines, if followed, help in preventing errors, controlling the complexity of the program, and increasing the readability and understandability of the program. A set of comprehensive coding guidelines encompasses all aspects of code development. To ensure that all developers work in a harmonized manner (the source code should reflect a harmonized style as a single developer had written the entire code in one session), the developers should be aware of the coding guidelines before starting a software project. Moreover, coding guidelines should state how to deal with the existing code when the software incorporates it or when maintenance is performed.

Since there are numerous programming languages for writing software codes, each having different features and capabilities, coding style guidelines differ from one language to another. However, there are some basic guidelines which are followed in all programming languages. These include naming conventions, commenting conventions, and formatting conventions.

There are certain rules for naming variables, functions and methods in the software code. These naming conventions help software developers in understanding the use of a particular variable or function. The guidelines used to assign a name to any variable, function, and method are listed below.

1. All the variables, functions, and methods should be assigned names that make the code more understandable to the reader. By using meaningful names, the code can be self-explanatory, thus, minimizing the effort of writing comments for variables. For example, if two variables are required to refer to 'sales tax' and

'income tax', they should be assigned names such as 'sales Tax' and 'income Tax'.

2. For names, a full description in a commonly spoken language (for example, English) should be used. In addition, the use of abbreviations should be avoided. For example, variable names like 'contact Number' and 'address' should be used instead of 'cno' and 'add'.

3. Short and clear names should be assigned in place of long names. For 'example, 'multiply The Two Numbers' can be shortened to 'multiply Numbers' as it is clear and short enough to be expressed in reasonable length.

In every programming language, there is a different naming convention for variables and constants in the software code. The commonly used conventions for naming variables and constants are listed in Table.

Variable Naming Conventions

Constant Naming Conventions

The variable names should be in camel case letters starting with a lower-case letter. For example, use 'total Amount' instead of 'Total Amount'. All the names of constants should be in upper case. In case the name of constant is too long, it should be separated by an underscore. For example, sales tax rate should be written as 'SALES_TAX_RATE'.

The temporary storage variables that are restricted to a segment of code should be short. For example, the variable 'temp' can be used for a temporary variable. It is important to note that a single temporary variable should not be reused in the same program. For example, variables 'i', 'j', or 'k' are declared while using loops. The use of literal should be avoided. Literal numbers such as '15' used in the software code confuses the reader. These numbers are counted as integers and result in wrong output of the program. However, the numbers '0' and '1' can be used as constants.

The use of numbers in naming variables should be avoided. For example, 'First Number' should be used instead of 'Number

As with variables and constants, there are some guidelines that should be followed while naming functions in the software code. These conventions are listed below.

- The names of functions should be meaningful and should describe the purpose of the function with clarity and briefness. Like variables, the names should be self-explanatory so that no additional description about the task of that function is required.

- The function name should begin with a verb. For example, the verb 'display' can be used for the function that displays the output on the screen. In case the verb itself is not descriptive, an additional noun or adjective can be used with the verb. For example, the function name 'add Marks' should be used to clarify the function and its purpose.

- In case the function returns a Boolean value, the helping verbs 'is' and 'has' should be used as prefixes for the function name. For example, the function name 'is Deposited' or 'has Deposited' should be used for functions that return true or false values.

Comments are helpful in proper understanding of the code segment used in program. Commenting conventions should be used efficiently to make the code easy to grasp. Generally, two types of commenting conventions are used: file header comments and trailing comments.

file header comments

Python is a suitable language for scriptwriters and developers. Let's write a script for Voice Assistant using Python. The query for the assistant can be manipulated as per the user's need.

Speech recognition is the process of converting audio into text. This is commonly used in voice assistants like Alexa, Siri, etc. Python provides an API called SpeechRecognition to allow us to convert audio into text for further processing. In this article, we will look at converting large or long audio files into text using the SpeechRecognition API in python.

Modules needed

Sub process:- This module is used to get system subprocess details used in various commands i.e Shutdown, Sleep, etc. This module comes built-in with Python.

Wolfram Alpha:- It is used to compute expert-level answers using Wolfram's algorithms, knowledgebase and AI technology. To install this module type the below command in the terminal. `pip install wolframalpha`

Pytttsx3:- This module is used for the conversion of text to speech in a program it works offline. To install this module type the below command in the terminal. pip install pyttsx3

Tkinter:- This module is used for building GUI and comes inbuilt with Python. This module comes built-in with Python.

Wikipedia:- As we all know Wikipedia is a great source of knowledge just like GeeksforGeeks we have used the Wikipedia module to get information from Wikipedia or to perform a Wikipedia search. To install this module type the below command in the terminal. pip install Wikipedia

Speech Recognition:- Since we're building an Application of voice assistant, one of the most important things in this is that your assistant recognizes your voice (means what you want to say/ ask). To install this module type the below command in the terminal. pip install SpeechRecognition

Web browser:- To perform Web Search. This module comes built-in with Python.

Ecapture:- To capture images from your Camera. To install this module type the below command in the terminal. pip install ecapture

Pyjokes:- Pyjokes is used for the collection of Python Jokes over the Internet. To install this module type the below command in the terminal. pip install pyjokes

Datetime:- Date and Time are used to showing Date and Time. This module comes built-in with Python.

Twilio:- Twilio is used for making calls and messages. To install this module type the below command in the terminal. pip install twilio
Requests: Requests is used for making GET and POST requests. To install this module type the below command in the terminal. pip install requests

BeautifulSoup: BeautifulSoup is a library that makes it easy to scrape information from web pages. To install this module type the below command in the terminal. pip install beautifulsoup4

Note: You can remove some of the import files if you don't want to get that feature as here Twilio for making calls and messages if you don't want to use that you can simply remove that function.

Implementation

Import the below libraries.

```
import subprocess
```

```
import wolframalpha
```

```
import pyttsx3
```

```
import tkinter
```

```
import json
```

```
import random
```

```
import operator
```

```
import speech_recognition as sr
```

```
import datetime
```

```
import wikipedia
```

```
import webbrowser
```

```
import os
```

```
import winshell
```

```
import pyjokes
```

```
import feedparser
```

```
import smtplib
```

```
import ctypes
```

```
import time
```

```
import requests
```

```
import shutil
```

```
from twilio.rest import Client

from clint.textui import progress

from ecapture import ecapture as ec

from bs4 import BeautifulSoup

import win32com.client as wincl

from urllib.request import urlopen
```

TRAILING COMMENTS

Are used to provide explanation of a single line of code. These comments are used to clarify the complex code. These also specify the function of the abbreviated variable names that are not clear. In some languages, trailing comments are used with the help of a double slash (/). The commenting conventions that are commonly followed in the software code are listed below.

- Comments should not be used to include information that is clearly understandable from the

SOFTWARE

- Comments should be used with important segments of code and code segments that are difficult to understand.
- Comments should be separated from the code to enhance readability of the Software Code.

Formatting (way of arranging a program in order to enhance readability) consists of indentation, alignment, and use of white spaces in the program. Consistency plays an important role while formatting a program in an organized way. A program with consistent formatting makes the code easier to read and understand. The commonly used formatting conventions are listed below.

INTENTATION

This refers to one or more spaces left at the beginning of statements in the program. Indentation is useful in making the code easily readable. However, the spaces used for indentation should be followed in the entire program. The guidelines that are commonly followed while indenting a program are listed below.

WHITE SPACES:

These improve readability by minimizing the compactness of the code. Some of the guidelines for proper usage of spaces within the code are listed below.

- Indentation should be used to highlight a nested block. Some nested blocks can be made with the help of 'if-else' and 'do-while' loops.
- Indentation is required if the statement is large enough to fit in a single line.
- Indentation should be consistent at the beginning and at the end of the braces in the program.
- There should be a space after placing a comma between two function arguments.
- There should be no space between a function name and parenthesis.
- There should be spaces to align the operators vertically to emphasize program structure and semantics.

If coding guidelines are used in a proper manner, errors can be detected at the time of writing the software code. Such detection in early stages helps in increasing the performance of the software as well as reducing the additional and unplanned costs of correcting and removing errors.

Implementing coding guidelines Moreover, if a well-defined coding guideline is applied, the program yields a software system that is easy to comprehend and maintain. Some of the coding guidelines that are followed in a programming language are listed below.

- All the codes should be properly commented before being submitted to the review team.
- All curly braces should start from a new line.
- All class names should start with the abbreviation of each group. For example, AA and CM can be used instead of academic administration and course management, respectively.
- Errors should be mentioned in the following format: [error code]: [explanation]. For example, 0102: null pointer exception, where 0102 indicates the error code and null pointer exception is the name of the error.
- Every 'if' statement should be followed by a curly brace even if there exists only a single statement.

- Every file should contain information about the author of the file, modification date, and version information.

Similarly, some of the commonly used coding guidelines in a database (organized collection of information that is systematically organized for easy access and analysis) are listed below.

- Table names should start with TBL. For example, TBL_STUDENT.
- If table names contain one word, field names should start with the first three characters of the name of the table. For example, STU_FIRSTNAME.
- Every table should have a primary key.
- Long data type (or database equivalent) should be used for the primary key

ADVATAGES OF CODING GUIDLINES

Coding guidelines supplement the language standard by defining acceptable and unacceptable usage of the programming language used. Acceptable usage avoids troublesome situations while unacceptable usage is conducive to errors or leads to misunderstanding of the written code. Properly implemented coding guidelines help the developer to limit program complexity, establish the basis for code review, and guard against compiler and common programming errors. Other advantages associated with coding guidelines are listed below and depicted.

TESTING

INTRODUCTION

Testing is the process of evaluating a system, application, or software product to identify any defects or errors and to verify that it meets the specified requirements and expected behavior. Testing is an essential part of the software development life cycle, as it helps to ensure the quality and reliability of the software product.

THE MAIN OBJECTIVES OF TESTING ARE TO:

Verify that the system or application works as intended and meets the specified requirements.

Identify any defects, errors, or inconsistencies in the system or application.

Improve the overall quality of the system or application.

Ensure that the system or application is reliable and performs well under different conditions.

Provide feedback to the development team to improve the design and functionality of the system or application.

There are various types of testing, including functional testing, performance testing, security testing, usability testing, and more. Each type of testing focuses on specific aspects of the system or application and helps to ensure that it meets the desired quality standards. Testing can be conducted manually or through the use of automated testing tools and frameworks.

Overall, testing is a critical aspect of software development and helps to ensure that the software product is of high quality and meets the needs and expectations of its users.

TESTING OBJECTIVES

The main objectives of testing are:

To verify that the system or application works as intended and meets the specified requirements.

To identify any defects, errors, or inconsistencies in the system or application.

To improve the overall quality of the system or application.

To ensure that the system or application is reliable and performs well under different conditions.

To provide feedback to the development team to improve the design and functionality of the system or application.

LET'S LOOK AT EACH OBJECTIVE IN MORE DETAIL:

Verification: The first objective of testing is to verify that the system or application works as intended and meets the specified requirements. This involves testing the system or application to ensure that it performs the functions it is supposed to and that it meets the expected behavior and standards.

Defect identification: Testing also aims to identify any defects, errors, or inconsistencies in the system or application. By identifying these issues, the development team can address and fix them, ensuring that the system or application is of high quality and functions as expected.

Quality improvement: The goal of testing is to improve the overall quality of the system or application. Testing helps to ensure that the system or application is reliable, robust, and free of defects, providing a high-quality product to the end-users.

Reliability: Testing helps to ensure that the system or application is reliable and performs well under different conditions. This includes testing the system or application in various environments and situations to ensure that it can handle different scenarios.

Feedback: Testing provides feedback to the development team to improve the design and functionality of the system or application. This feedback helps the team to identify areas where improvements can be made and make necessary changes to ensure that the system or application meets the needs and expectations of its users.

TESTING STRATEGIES

There are several testing strategies that can be used to ensure the quality of a system or application. Some of the commonly used testing strategies include:

Black-box testing: In black-box testing, the internal workings of the system or application are not considered, and testing is done based on the input/output behavior of the system or application. The aim of this testing is to ensure that the system or application meets the specified requirements and performs the expected functions.

White-box testing: In white-box testing, the internal workings of the system or application are considered, and testing is done based on the internal code and logic of the system or application. The aim of this testing is to identify any defects or errors in the code and ensure that the system or application is functioning as expected.

Regression testing: Regression testing is used to ensure that the changes made to the system or application do not have any adverse effects on the existing functionality. This testing is done after each change to the system or application to ensure that all existing features continue to work as expected.

Acceptance testing: Acceptance testing is done to ensure that the system or application meets the requirements of the end-users or customers. This testing is typically done by the end-users or customers themselves, and it ensures that the system or application is usable and acceptable to its intended users.

Exploratory testing: Exploratory testing is done to explore the system or application and identify any defects or issues that may not be covered by other testing strategies. This testing is typically done by experienced testers who are familiar with the system or application and can identify any issues that may have been missed in other testing strategies.

Performance testing: Performance testing is done to test the system or application's performance under various conditions, such as load testing, stress testing, and endurance testing. The aim of this testing is to ensure that the system or application performs well under different conditions and can handle the expected load.

Overall, the testing strategy used will depend on the specific requirements of the system or application, as well as the development team's preferences and expertise. By using a combination of different testing strategies, the development team can ensure that the system or application is of high quality and meets the needs and expectations of its users.

TYPES OF TESTING

For electronic parts:

- power supply voltages and currents
- signal levels and frequencies at various test points,
- range of operation to check linearity and accuracy, and so on.

For mechanical parts:

Dimensional tolerances:

Dimensional tolerances are a crucial part of the design and manufacturing process. A tolerance is a numerical range of measurements assigned to a part's dimensions indicating how much a manufacturing team can drift from the nominal measurement.

range of motion (i.e., speed, distance)

Apart from physical examination, ROM evaluation is most frequently performed using a device called a goniometer, an instrument that measures the angle at a joint. Goniometers measure range of motion in degrees, from zero to 180 or 360, and come in various shapes and sizes to be used on particular joints.

Forces:

To check the forces acting on a mechanical part, you can follow the steps below:

Identify the external loads: The first step is to identify all the external loads that are acting on the mechanical part. This can include forces, moments, and torques, as well as any other types of loads that may be present.

Determine the reaction forces: Once you have identified the external loads, you need to determine the reaction forces at the supports of the mechanical part. This can be done by applying the principles of statics and solving for the unknown forces using the equations of equilibrium.

Analyze the internal forces: With the external loads and reaction forces identified, you can then analyze the internal forces within the mechanical part. This can be done using the principles of mechanics, such as stress and strain analysis, as well as finite element analysis (FEA) or other simulation tools.

Check for failure: Finally, you should check the internal forces against the material properties of the mechanical part to ensure that it is not experiencing excessive stress or deformation. If the internal forces exceed the material's strength, the part may fail, and you will need to redesign it or modify the loading conditions.

Overall, checking the forces in a mechanical part requires a thorough understanding of mechanics and the use of appropriate analytical and simulation tools. It's important to perform this analysis to ensure that the mechanical part can withstand the external loads it will experience during its intended use.

Temperatures:

To check the temperatures in a mechanical part, you can follow the steps below:

Identify the potential sources of heat: The first step is to identify the potential sources of heat that can affect the mechanical part. This can include external factors such as the environment in which the part is located, as well as internal factors such as friction and heat generated by machinery.

Measure the temperatures: Once you have identified the potential sources of heat, you need to measure the temperatures at various points on the mechanical part. This can be done using temperature sensors such as thermocouples, resistance temperature detectors (RTDs), or infrared cameras.

Analyze the temperature data: With the temperature data collected, you can analyze it to determine the temperature distribution within the mechanical part. This can be done using data analysis tools such as spreadsheets, data visualization software, or specialized temperature analysis software.

Check for overheating: Finally, you should check the temperature data against the maximum operating temperature of the mechanical part and the temperature limits of the materials it is made of. If the temperatures exceed these limits, the part may be at risk of overheating and failure, and you may need to modify its design or operating conditions.

Overall, checking the temperatures in a mechanical part requires a thorough understanding of heat transfer and the use of appropriate temperature measurement and analysis tools. It's important to perform this analysis to ensure that the mechanical part can operate safely and efficiently within its intended temperature range.

and on and on, because there are so many types of mechanical parts.

For optical parts:

Power input and output:

To check the power input and output in mechanical parts, you can follow the steps below:

Identify the power source and load: The first step is to identify the power source that supplies power to the mechanical part and the load that the mechanical part drives.

Measure the power input: Once you have identified the power source, you need to measure the power input to the mechanical part. This can be done using power meters, current transformers, or other power measurement devices.

Measure the power output: After measuring the power input, you need to measure the power output of the mechanical part. This can be done using dynamometers, torque sensors, or other devices that measure mechanical power output.

Analyze the power data: With the power data collected, you can analyze it to determine the efficiency of the mechanical part. This can be done by calculating the power output divided by the power input and expressing the result as a percentage. A high percentage indicates a more efficient mechanical part, while a low percentage indicates a less efficient mechanical part.

Check for power losses: Finally, you should check for power losses within the mechanical part. Power losses can occur due to friction, mechanical losses, or other factors that reduce the efficiency of the mechanical part. If power losses are excessive, you may need to modify the design or operating conditions of the mechanical part to improve its efficiency.

Overall, checking the power input and output in mechanical parts requires a thorough understanding of power measurement and analysis and the use of appropriate measurement devices. It's important to perform this analysis to ensure that the mechanical part operates efficiently and effectively within its intended application.

For communications parts:

bandwidths:

To check the bandwidth in communication parts, you can follow the steps below:

Define the communication system: The first step is to define the communication system, including the transmitter, receiver, and communication channel. This includes the type of modulation and encoding used to transmit information.

Generate a test signal: Once the communication system is defined, you need to generate a test signal that covers the full bandwidth of the system. This can be done using a function generator, signal generator, or other test equipment.

Measure the frequency response: With the test signal generated, you need to measure the frequency response of the communication system. This can be done using a spectrum analyzer, network analyzer, or other frequency measurement equipment. The frequency response shows how the system responds to different frequencies within its bandwidth.

Calculate the bandwidth: After measuring the frequency response, you can calculate the bandwidth of the communication system. The bandwidth is typically defined as the range of frequencies over which the system can transmit information at a specific level of performance, such as a certain level of signal-to-noise ratio (SNR) or bit error rate (BER).

Check for system limitations: Finally, you should check for any limitations in the communication system that may affect its bandwidth. This can include factors such as noise, distortion, or other sources of interference that can degrade the performance of the system. If these limitations are significant, you may need to modify the system design or operating conditions to improve its bandwidth.

Overall, checking the bandwidth in communication parts requires a thorough understanding of communication systems and the use of appropriate measurement equipment. It's important to perform this analysis to ensure that the communication system can transmit information at the desired level of performance within its available bandwidth.

Transmission power:

To check the transmission power in communication parts, you can follow the steps below:

Identify the communication system: The first step is to identify the communication system, including the transmitter and receiver, as well as the type of communication technology being used (e.g. wireless, wired, etc.).

Measure the output power: Once the communication system is identified, you need to measure the output power of the transmitter. This can be done using a power meter or other power measurement device, which is connected to the output port of the transmitter.

Calculate the transmission power: After measuring the output power, you can calculate the transmission power of the communication system. The transmission power is typically

defined as the power level at which the transmitter sends signals to the receiver. This can be calculated by subtracting any losses that occur due to factors such as attenuation or interference.

Check for compliance with regulations: Finally, you should check that the transmission power of the communication system complies with regulatory requirements. Depending on the type of communication technology being used, there may be limits on the maximum transmission power allowed. For example, in wireless communications, regulatory bodies such as the Federal Communications Commission (FCC) in the US, specify maximum transmission power limits for different frequency bands.

Overall, checking the transmission power in communication parts requires a thorough understanding of communication systems and the use of appropriate power measurement equipment. It's important to perform this analysis to ensure that the transmission power of the communication system is at the appropriate level to achieve the desired communication range and to comply with regulatory requirements.

Receive power:

To check the receive power in communication parts, you can follow the steps below:

Identify the communication system: The first step is to identify the communication system, including the transmitter and receiver, as well as the type of communication technology being used (e.g. wireless, wired, etc.).

Measure the input power: Once the communication system is identified, you need to measure the input power of the receiver. This can be done using a power meter or other power measurement device, which is connected to the input port of the receiver.

Calculate the receive power: After measuring the input power, you can calculate the receive power of the communication system. The receive power is typically defined as the power

level at which the receiver receives signals from the transmitter. This can be calculated by subtracting any losses that occur due to factors such as attenuation or interference.

Check for compliance with performance specifications: Finally, you should check that the receive power of the communication system meets the required performance specifications. The required receive power depends on the type of communication technology being used and the specific application. For example, in wireless communications, the required receive power may depend on factors such as the distance between the transmitter and receiver, the modulation scheme being used, and the bit error rate (BER) required for the communication.

Overall, checking the receive power in communication parts requires a thorough understanding of communication systems and the use of appropriate power measurement equipment. It's important to perform this analysis to ensure that the receive power of the communication system is at the appropriate level to achieve the desired communication performance.

Bit-error-rates:

Bit Error Rate (BER) is the measure of the rate at which errors occur in a digital communication system. It is defined as the ratio of the number of bits that are incorrectly received to the total number of bits transmitted during a specific time interval.

In other words, BER is a measure of the quality of the digital transmission channel, and it indicates the probability that a transmitted bit will be received in error. A higher BER indicates a lower quality of the channel, while a lower BER indicates a higher quality of the channel.

BER is typically expressed as a decimal fraction or a percentage. For example, a BER of 10^{-6} means that one bit in a million is received incorrectly, while a BER of 10^{-9} means that one bit in a billion is received incorrectly. Lower BER values indicate better transmission quality, while higher BER values indicate poorer transmission quality.

Distortion:

deviation from an ideal or that occurs during transmission, procesIn general, distortion refers to any sing, or reproduction. In the context of communications and signal processing, distortion is any change that occurs to a signal as it passes through a system desired signal or medium, and it can occur in many forms.

Some common types of distortion include:

Amplitude distortion: This occurs when the amplitude of a signal is changed during transmission or processing. For example, if a signal's amplitude is reduced due to attenuation in a communication channel, it can be considered amplitude distortion.

Phase distortion: This occurs when the phase of a signal is changed during transmission or processing. Phase distortion can cause signals to be shifted in time, which can affect the quality of the signal.

Frequency distortion: This occurs when the frequency content of a signal is changed during transmission or processing. Frequency distortion can cause certain frequencies to be attenuated or amplified more than others, which can alter the spectral characteristics of the signal.

Nonlinear distortion: This occurs when the relationship between the input and output of a system is nonlinear. Nonlinear distortion can cause harmonics and other unwanted signal components to be generated.

Inter modulation distortion: This occurs when two or more signals are combined in a nonlinear system, causing new frequencies to be generated. Intermodulation distortion can cause interference and reduce the quality of the signals.

Overall, distortion can have a significant impact on the quality of signals in communications and signal processing applications. It's important to understand the types of distortion that can occur and take steps to minimize their effects on signal quality.

Unit testing

Unit testing is a software testing technique in which individual units or components of software are tested in isolation from the rest of the application to ensure that they meet their functional requirements and produce expected outputs. The goal of unit testing is to identify and fix defects early in the software development cycle, before the components are integrated into larger systems.

In unit testing, the individual units of code are tested using automated test cases that are designed to exercise different scenarios and edge cases. The tests are typically written by the software developers themselves and are run on a regular basis as part of the development process.

Some of the benefits of unit testing include:

Early detection of defects: Unit testing allows developers to catch defects early in the development cycle, when they are easier and cheaper to fix.

Improved code quality: Unit testing can help improve the quality of the code by identifying and eliminating defects before they are integrated into the larger application.

Faster debugging: When defects are found during unit testing, it is easier to identify the source of the problem and fix it quickly.

More reliable software: Unit testing can help ensure that software components are reliable and perform as expected, which can improve the overall quality of the application.

Unit testing is a key part of modern software development practices and is often used in conjunction with other testing techniques, such as integration testing and system testing. By catching defects early and ensuring that individual components work as expected, unit testing can help produce high-quality software that meets the needs of users.

Integration testing

Integration testing is a software testing technique that is used to verify the functionality of different modules or components of a software system when they are combined and interact with each other. The goal of integration testing is to ensure that the integrated system works as expected and that the individual components work correctly together.

In integration testing, different modules or components of the software system are combined and tested as a group. The tests are designed to exercise the interactions between the modules and to ensure that the data is passed correctly between them. Integration testing can be performed at different levels of granularity, including module-level integration testing, subsystem-level integration testing, and system-level integration testing.

Some of the benefits of integration testing include:

Early detection of defects: Integration testing can help detect defects early in the development cycle, before the integrated system is released to users.

Improved quality of the integrated system: Integration testing can help ensure that the integrated system works correctly and that the different modules or components interact as expected.

Increased confidence in the integrated system: Integration testing can help build confidence in the integrated system and ensure that it is reliable and performs as expected.

Reduced risk: By verifying the functionality of the integrated system, integration testing can help reduce the risk of defects and failures when the system is deployed to users.

Overall, integration testing is an important part of the software development process and is essential for ensuring that the integrated system works correctly and meets the needs of users

System testing

System testing is a software testing technique that is used to verify that a software system meets its functional and non-functional requirements and works as intended. The goal of system testing is to evaluate the system's overall performance, reliability, security, and usability under different conditions and scenarios.

In system testing, the software system is tested as a whole, rather than testing individual components or modules. System testing is typically performed after integration testing and before the system is released to users. The tests are designed to simulate real-world usage scenarios and to ensure that the system works as expected in different environments and with different configurations.

Some of the benefits of system testing include:

Ensuring the system meets requirements: System testing helps verify that the system meets its functional and non-functional requirements and that it works as intended.

Improving system quality: System testing helps identify and fix defects and issues that could impact the performance, reliability, security, or usability of the system.

Reducing risk: By verifying the functionality and performance of the system, system testing helps reduce the risk of defects and failures when the system is deployed to users.

Building user confidence: System testing can help build user confidence in the system by ensuring that it performs reliably and as expected under different scenarios.

Overall, system testing is an important part of the software development process and is essential for ensuring that the software system works correctly and meets the needs of users.

Validation testing

Validation testing is a type of software testing that aims to verify that a system or application meets the specifications and requirements of the user or customer. This testing is done to ensure that the system or application is working as intended and is free from defects or errors.

Validation testing is usually conducted after the development of the system or application is complete and is typically done in a production-like environment. It is used to verify that the system or application is ready for release to the end-users or customers.

Some common techniques used in validation testing include functional testing, regression testing, integration testing, and acceptance testing. The goal of validation testing is to ensure that the system or application meets the needs of the users and customers, and that it is reliable and robust enough to handle various use cases and scenarios.

User acceptance test (UTA)

User acceptance testing (UAT) is a type of testing that is conducted to ensure that a system or application meets the requirements and needs of its end-users. In other words, UAT is used to verify that the system or application is usable and acceptable to its intended users.

UAT is typically conducted by the end-users or customers themselves, rather than the development team or testers. This helps to ensure that the system or application meets the needs and expectations of the actual users, rather than just meeting technical specifications.

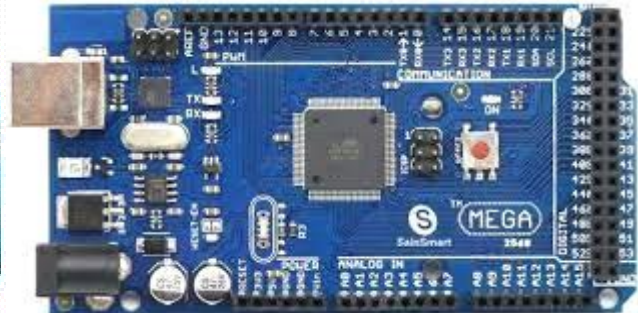
The process of UAT typically involves creating test scenarios and cases that simulate real-world usage of the system or application. The end-users or customers then execute these test scenarios and provide feedback on any issues or areas where improvements are needed. The feedback is then analyzed by the development team, and any necessary changes or improvements are made.

The goal of UAT is to ensure that the system or application is usable and acceptable to its end-users or customers, and that it meets their needs and expectations. By conducting UAT, organizations can help to ensure that their systems and applications are successful and well-received by their users.

Pictures of hardware components used



Arduino uno



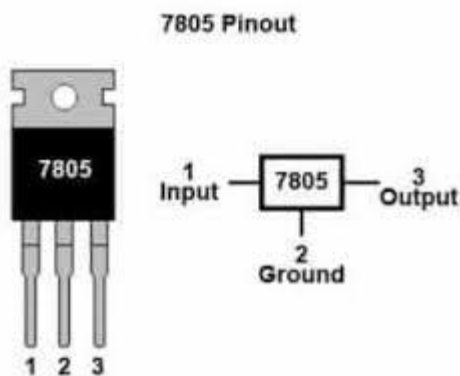
Arduino mega



hc05 bluetooth module



mg995 servo motor



voltage regulating ic



12 volt 30rpm motor



12v 12ah battery



12v fuse



switch



8ohms speaker



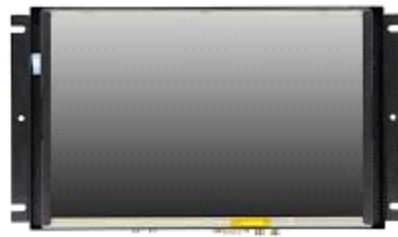
5v class-D amplifier



3.5mm aux pin



ir sensor



7-inch screen



DHT-11



MAX30100



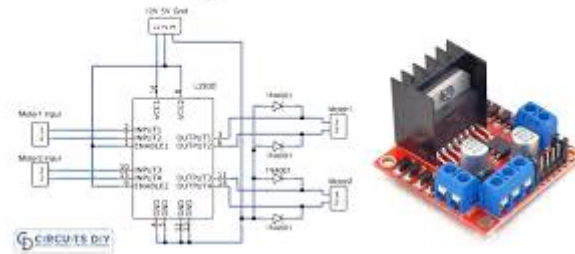
voice sensor



12-v battery charger



H Bridge Motor Driver



usb socket

12v motor driver

CONCLUSION AND FUTURE ENHANCMENT

FUTURE ENHANCEMENT

- . check the humidity level
- . Act as voice assistant
- . Can control movment using app
- . Line following robot
- . Check the humidity leve
- . Shows the readings in display

CONCLUSION

In this research we are trying to make an advance type of robot which can interact as a human as well as a personal health care companion which can be upgraded into much more in the future. The applications in this field are limitless. we can also upgrade this robot into self serving robot, security robot etc